

Donghyung Lee

DevOps Engineer · 4Y 3M Experience

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Introduction

Hello, I am Donghyung Lee, a DevOps Engineer.

Aiming for stable infrastructure operation, I codify every change as IaC and turn incident response into postmortem-driven assets to cut cloud costs 20-50% while raising operational efficiency. My core strength is interrogating the purpose of each task and pursuing a better direction.

I design and build infrastructure across AWS and GCP cloud and an OpenStack-based on-premises PaaS. I standardize container operations with Kubernetes, codify infrastructure with Terraform and Ansible, and run GitLab CI / ArgoCD-centered deploy pipelines alongside monitoring and logging stacks. Repetitive work is automated in Python and Bash.

I also operate adjacent data-engineering and AI/ML ops through data-pipeline automation and an in-house GPU LLM service, and build in-house DevOps tools myself — git-bridge, a Slack APK bot, and more.

Across 3 companies I solely led AWS → GCP migration, AWS + on-premises hybrid, and on-premises PaaS builds — achieving 20% AWS and 50% GCP cost cuts, 50-67% faster deploys, and 90%+ manual-work automation. Most recently I built an AWS EKS multi-cluster for a new game's cloud production launch, migrated helmfile-push deploys to a more stable auto-syncing model (ArgoCD pull-GitOps), and — alongside resolving 14-second game-DB stalls and unifying login via Keycloak central SSO — raised operational efficiency another notch.

I aim to keep contributing to infrastructure architecture advancement and organizational productivity.

Strengths

Asking 'Why' First

- Habit of asking 'why' first — define the business problem before picking the tool

Systematic Documentation & Knowledge Sharing

- Document and share every task, building organizational knowledge assets

Record-Based Continuous Improvement

- Postmortem-record every incident, infra change, and troubleshooting — preventing recurrence, driving improvement

Full-Stack Infrastructure Experience

- End-to-end infra across cloud (AWS, GCP), on-premises, networking, CI/CD, and monitoring

Skills

Cloud	AWS, GCP, OpenStack
Container Orchestration	Kubernetes, Docker, Cilium CNI, NGINX Gateway Fabric / Gateway API, Karpenter
Infrastructure as Code	Terraform, Ansible, Helm, Helmfile
CI/CD	GitHub Actions, GitLab CI/CD, ArgoCD, ArgoCD ApplicationSet, Jenkins, GitOps
Monitoring & Logging	PLG Stack (Prometheus/Loki/Grafana), ELK Stack, EFK Stack, ECK Operator, Fluent Bit, Thanos
Security & IAM	Keycloak (Operator, OIDC IdP), Vaultwarden, GitLab SSO (OIDC), External Secrets Operator, Sealed Secrets
Automation	Go, Python, Shell Scripting
Database	MySQL, PostgreSQL, Redis, MongoDB
OS & Servers	Ubuntu, Rocky Linux, CentOS, Debian, GPU Server

Professional Experience

Phantom (Concrit Studio)

2024.10 ~ Present

DevOps Engineer

Sole DevOps infrastructure engineer at a game studio, designing, building, and operating AWS-based cloud and on-premises servers. Codified all infrastructure with Terraform/Ansible (100% IaC for new components) and documented operations via automation scripts and postmortems, keeping a hand-off-ready setup despite solo ownership. Also achieved both cost optimization and operational stability through hybrid cloud architecture.

- Cut monthly infra cost \$60,000 → \$48,000 (20%) — owned AWS + on-prem hybrid design end-to-end, 100% Terraform/Ansible IaC for new components (IAM excluded), role-separated envs for fault isolation, partnering with the external publisher's ops team on tier-2 support
- Cloud production-launch foundation for a new game — expanded a single on-prem dev cluster into an on-prem + AWS production multi-cluster (new EKS build), 11 components 100% GitOps (ArgoCD), chose Karpenter over Cluster Autoscaler to unify & autoscale 9 spot families (faster provisioning, cost efficiency), keyless deploy via IRSA/Pod Identity · SSM (zero SSH keys) removing leak risk
- In-house K8s monorepo (25 components) CI/CD & automation — CI first-job bootstrap 5min → ~10s (15 repos, shared toolchain images), GitOps wave-based auto-deploy (GitLab CI · ArgoCD · Jenkins), shell → Python (stdlib only) automation migration (8 canonical templates · 758 unittests)
- helmfile push → ArgoCD pull-GitOps migration (24 releases / 2 clusters, zero-downtime adoption), risk-tiered (T1/T2/T3) dependency supply chain with auto-MR, Kubespray K8s upgrade automation (etcd backup · health checks)
- Observability + product analytics — 3-stage logging redesign ELK → EFK → ECK Operator (Fluent Bit HA · SQLite offset DB for zero log gap/duplication on Pod restart), kube-prometheus-stack (alerts · dashboards · 4 DB exporters · Cilium) for full-stack visibility, extended the EFK stack into a 12-panel Kibana analytics platform via ES Transform cohort (user-retention D+1~D+30), \$0 SaaS
- Zero-downtime migration of multiple ingress-nginx instances to NGINX Gateway Fabric (single control-plane, per-class Gateway CRs, wildcard TLS) — internal standardization artifacts released as 3 reusable OSS Helm charts on ArtifactHub
- Central SSO + GitOps secret management — Keycloak Operator unifying Harbor · ArgoCD · Vaultwarden OIDC (login with existing GitLab accounts, zero downtime, 70 users / 50 secrets), External Secrets Operator removing plaintext DB passwords, Harbor pull secret as a SealedSecret
- Built & open-sourced internal productivity tools — Slack APK bot (90% faster QA delivery), GitLab ↔ Google Drive sync (100/wk), internal LLM (Ollama + Open WebUI, drove onboarding for 15 devs, replacing \$100/mo SaaS), Go OSS git-bridge / static-file-server

Key Achievements

- 20% Monthly Infra Cost Reduction (\$60,000 → \$48,000, ~\$144K annual savings → reinvested in new projects)
- 67% Deployment Time Reduction (30min → 10min, 3x weekly deployments for faster market response)
- 14s → 0 Ended Game DB Stalls (etcd fsync 454x faster: 1760ms → 3.88ms, control-plane SSD + GitLab Runner I/O isolation VM)
- 95% CI Bootstrap Time Reduction (Layered shared toolchain images, 5min → ~10s, across 15 repos)
- 90% Log Search Time Reduction (per-Pod kubectrl exec + grep avg 10min → single Kibana interface 1min)

NerdyStar

2023.03 ~ 2024.07

DevOps Engineer

Led the large-scale AWS → GCP cloud migration at a game and blockchain startup, and built GitOps-based CI/CD pipelines.

- AWS → GCP large-scale migration (Shared VPC + Workload Identity Federation keyless GitHub Actions auth, subdomain-delegated DNS cutover during a maintenance window, Cloud Armor + Cloud CDN, self-hosted NFS replacing Filestore)
- Built GitOps CI/CD pipeline on GitLab CI · ArgoCD
- Automated certificates with Cert-Manager + Let's Encrypt
- Built game/blockchain data pipeline (MongoDB · CloudSQL · GA · Dune → BigQuery → Google Sheets, Cloud Functions + Scheduler + Dataflow, Terraform IaC)
- Built integrated monitoring on PLG Stack (Prometheus, Loki, Grafana)
- Designed & built GPU server infrastructure for internal LLM services

Key Achievements

- 50% Cloud Cost Reduction (\$10,000 → \$5,000/month)
- 50% Deployment Time Reduction (40min → 20min)
- 95% Rollback Time Reduction (30min → 1-2min)
- 95% Data Collection Automation (0% data gap rate)
- 3x Deployment Frequency (1-2 → 5-7 per week)

Iorchard

2022.04 ~ 2023.03

Infra Engineer

Built PaaS (Kubernetes + OpenStack + Ceph) infrastructure solutions for clients and provided technical support. Designed and built on-premises cloud infrastructure for financial and public sector clients, and led operations training.

- Customized PaaS solution design and construction (Kubernetes HA + OpenStack + Ceph)
- DP (Delivery Partner) Kubernetes operations training program design and delivery (6 months)
- Kubernetes certificate validity extension automation script (1yr → 10yr)
- OpenStack and Ceph storage cluster operations and optimization
- Regular cluster inspections and troubleshooting
- Ansible-based server provisioning automation eliminating configuration drift issues

Key Achievements

- 5 Cases Client PaaS Solution Deployments (Stable HA-based operation)
- 10x Certificate Renewal Cycle Extension (Annual → Once per 10 years)
- 50hrs Annual Ops Savings (Across 10 clients)

ETech System

2022.01 ~ 2022.04

System Engineer

- Hardware Server and SAN storage construction
- Transitioned to DevOps field for career growth

Opensource Projects

Kubernetes CRD

[k8s-namespace-sync](#) — Automatically sync Kubernetes namespace resources across clusters

[helios-lb](#) — Custom load balancer controller for Kubernetes

[network-policy-generator](#) — Kubernetes NetworkPolicy auto-generation controller

Helm Charts

[nginx-gateway-cr](#) — Tenant-level resource Helm chart for NGINX Gateway Fabric

[elasticsearch-eck](#) — Elasticsearch CR Helm chart for ECK Operator

[kibana-eck](#) — Kibana CR Helm chart for ECK Operator

[certmanager-letsencrypt](#) — Let's Encrypt resources Helm chart for cert-manager

GitHub Actions

[Compress Decompress Action](#) — Compress/decompress files in GitHub Actions workflows

[Go Git Commit Action](#) — Go-based Git commit creation and push

[Multi Git Mirror Action](#) — Automate mirroring across multiple Git repositories

[Image Tag Updater](#) — Automatically update image tags in YAML or Helm files

[Kube Diff Action](#) — Compare and report Kubernetes resource changes

[Helm Chart Release Action](#) — One-shot Helm chart release (package, gh-pages publish, OCI push)

[Go Kubebuilder Test Action](#) — kubebuilder operator test + manifests drift check

[Helm OCI Push](#) — Push Helm charts to OCI registries (GHCR, ECR, Harbor, Quay)

[Helm Kustomize Lint Action](#) — Lint and validate Helm charts (yamllint, helm lint, template, kubeconform)

[Major Tag Action](#) — Auto-update major version tags (v1) to the latest semver release

[Go Changelog Generator](#) — Generate changelogs from Conventional Commits

Ansible Galaxy

[Ansible K8s IAC Tool](#) — Automated installation collection for Kubernetes and IaC tools

[Ansible Kubectrl Krew](#) — kubectrl plugin Krew management role

[Ansible User Management](#) — Linux user and SSH key management role

DevOps Tools

[bash-pilot](#) — AI-powered Bash command assistant CLI tool

[git-bridge](#) — Git repository mirroring and sync CLI tool

[static-file-server](#) — Go-based static file server with directory UI, file preview, search, ZIP batch download

[kube-diff](#) — CLI to diff Kubernetes manifests (YAML, Helm, Kustomize) against a live cluster

[kube-events](#) — CLI to view and summarize Kubernetes events with grouping, filtering, colored output

Chrome Extensions

[Dev Toolkit](#) — A collection of useful developer utilities

[QRify](#) — Quickly generate QR codes for URLs or text

Education

Chungnam National University - Public Administration (Transfer)	2018.03 ~ 2020.08
National Institute for Lifelong Education - Business Administration (Academic Credit Bank)	2016.09 ~ 2018.02

Certifications

Engineer Information Processing	2021.11
AWS Certified Solution Architect - Associate (Expired)	2021.11
Network Administrator Level 2	2021.10
Linux Master Level 2	2021.10
Computer Specialist Level 1	2021.04
TOEIC Speaking Test - 130/Intermediate Mid 3 (Expired)	2021.02

Study

Tech blog (somaz.blog, English) — DevOps · Kubernetes · IaC writing for global readers	2025.01 ~ 현재
Tech blog (somaz.tistory.com, Korean) — ongoing DevOps · Kubernetes · IaC learning and operations notes	2023.04 ~ 현재
T101(Terraform) Study	2023.08 ~ 2023.10
AEWS(EKS) Study	2023.04 ~ 2023.06
PKOS(Kubernetes) Study	2023.01 ~ 2023.02
Cloud Security Training	2022.02 ~ 2022.04
Security Solution Operation Specialist Training Program	2021.07 ~ 2021.12